

Balance Scale Inverse Cards

Unit 7: Equation Detective Agency · Lesson 7-2 · EduWonderLab

UNIT 7 · LESSON 7-2 · TEACHER NOTES

Standard 6.EE.7

FORMAT Balance mat + solve

TIME 25–30 min

GROUPING Pairs

TOPIC Solve One-Step Addition and Subtraction Equations

STANDARD 6.EE.7

PREP LEVEL Light — print balance mats and equation cards

Learning goal *I can solve one-step addition and subtraction equations and check by substitution.*

Teacher setup

- Print one balance mat and card set per pair.
- Students use the inverse operation to solve, keeping the scale balanced.
- Every answer is checked by substituting back.

Materials

Balance mats, Equation cards, Inverse operation cards, Pencil.

Differentiation & language support

- Inverse operation cards: + undoes –, – undoes + (SPED).
- Sentence stem: “___ makes the equation true.”
- ESOL: show the same operation on both sides of the balance.

Key vocabulary (English / Español):

Term	Español	What it means
inverse operation	operación inversa	the operation that undoes another
solution	solución	the value that makes the equation true

Quick teacher check: *Require the substitution check on at least two cards.*

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Name: _____

Date: _____

Class: _____

Your goal: *I can solve one-step addition and subtraction equations and check by substitution.*

What you need

Balance mats, Equation cards.

How to play

1. Pick an equation card and place it on your balance mat.
2. Use the inverse operation on both sides.
3. Write the solution.
4. Check by substituting your answer back in.

Equation Cards

Use the inverse operation on both sides to solve. Then check by substituting.

1. $x + 5 = 12$	2. $x - 3 = 8$
3. $x + 9 = 15$	4. $x - 6 = 10$
5. $x + 12 = 20$	6. $x - 7 = 4$
7. $x + 4 = 4$	8. $x - 10 = 5$
9. $x + 2.5 = 6$	10. $x - 1/2 = 3$

Record your work (solve + check):

Explain your thinking

Explain why you do the same operation to both sides of an equation.

Sentence frame: *___ makes the equation true because when I substitute it, both sides equal ___.*

★ **Challenge:** Card 7 gives $x = 0$. Explain why a solution of zero still makes the equation true.

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. $x + 5 = 12$ Answer: $x = 7$	2. $x - 3 = 8$ Answer: $x = 11$
3. $x + 9 = 15$ Answer: $x = 6$	4. $x - 6 = 10$ Answer: $x = 16$
5. $x + 12 = 20$ Answer: $x = 8$	6. $x - 7 = 4$ Answer: $x = 11$
7. $x + 4 = 4$ Answer: $x = 0$	8. $x - 10 = 5$ Answer: $x = 15$
9. $x + 2.5 = 6$ Answer: $x = 3.5$	10. $x - 1/2 = 3$ Answer: $x = 3 \frac{1}{2}$

Teaching notes & look-fors

- Whatever you do to one side, do to the other to keep the balance.
- Card 7: $x + 4 = 4 \rightarrow x = 0$ is a valid solution.